

Experimental and numerical investigation of the aperture size effect on the efficient solar energy harvesting for solar thermochemical applications



J. Sarwar^a, G. Georgakis^b, K. Kouloulis^c, K.E. Kakosimos^{a,*}

^a Department of Chemical Engineering, Texas A&M University at Qatar, PO Box 23874, Doha, Qatar

^b Department of Mechanical Engineering, American University of Sharjah, Sharjah, United Arab Emirates

^c Department of Mechanical Engineering, Imperial College London, London SW7 2AZ, UK

ARTICLE INFO

Article history:

Received 20 October 2014

Accepted 22 December 2014

Keywords:

Heat transfer

Ray tracing

Variable irradiance

High flux solar simulator

ABSTRACT

In this paper, experimental and numerical work have been undertaken to investigate the steady state temperatures throughout the day of a cylindrical solar receiver when using fixed and variable size apertures. A high flux solar simulator, consisting of a 7 kW xenon short arc lamp, is employed as a light source. The sunlight intensity variations at early morning (06:30), morning (07:15) and noon (12:00) time of a reference day are imitated by changing the input current to the lamp. Experiments have been performed with different aperture diameters across selected irradiance levels to imitate sunlight variations. An optical model is developed to simulate incident flux distribution and the output is compared with the experimental measurements for validation. A finite volume algorithm is developed, based on a coupled Monte Carlo heat transfer model, to calculate the steady state temperatures in the receiver. Experimental and numerical temperatures are compared and an excellent agreement with an average temperature difference of $\pm 0.2\%$, is observed. The optimum aperture size varies with the change in irradiance intensity and therefore the time of day. Simulations for a 30 kW light source show that the daily steady state temperature differential for fixed apertures of 8–10 cm is 170–190 K. Variable apertures reduce power consumption by half when compared to fixed apertures. Variable apertures maintain steady state temperatures of 1000 K, 1100 K and 1200 K by consuming 26.8 kW day, 33.2 kW day and 26.9 kW day, respectively.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Solar energy is transient in nature due to the varying position of the sun and variable weather conditions, resulting in fluctuations in the incident flux density. This fluctuation inhibits the maintenance of even quasi-constant temperature in solar thermochemical applications [1] and negatively affects conversion efficiency. For example, in direct methane cracking a temperature variation in the range of 1148–1338 K results in hydrogen production variation in the range of ~ 5 –55%, [2]. Thus, a constant high wall temperature is desirable to achieve complete methane cracking [3]. Of fundamental importance to most solar reactors is the opening called the aperture. The aperture allows the incident concentrated solar energy to enter the reactor cavity, indirectly and directly heating the reactants and initiating thermochemical processes. Other parameters influencing the performance of the reactor are: optical

and re-radiation losses [4], conduction, convection, heat capacity and reactant feed conditions [5]. Optical losses and conduction depend on the reactor geometry, the material's absorptivity and its thermal conductivity. Re-radiation losses depend on reactor temperature and aperture size [4,6].

Extensive work has been carried out using fixed circular apertures with different reactor cavity geometries. For example, a rotating conical cavity-receiver reactor with a fixed aperture and a quartz window [5,7], a cylindrical cavity-receiver reactor with a tornado flow configuration [8], an insulated cylindrical cavity-receiver with a vortex flow [9], a tubular solar reactor with four tubular parallel graphite zones and a fixed circular aperture [10], and a cylindrical cavity-receiver reactor with a trumpet like lower part and a fixed circular aperture [11]. An ellipsoidal geometry of a receiver-reactor system with an absorption efficiency which is not very sensitive to aperture size has been proposed, [12,13]. Five different geometric configurations with fixed aperture size have been investigated numerically using the Monte Carlo ray tracing (MCRT) method at an incident irradiance of 1000 W m^{-2} [14] and with different aspect ratios of the cylindrical cavity receiver [15].

* Corresponding author.

E-mail address: k.kakosimos@qatar.tamu.edu (K.E. Kakosimos).

Nomenclature

A	area, m^2	t	thickness, m
C_p	heat capacity, $\text{J kg}^{-1} \text{K}^{-1}$	w	width of the back plate, m
h_c	convection heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$	<i>Greek letters</i>	
h_{out}	heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$	ε	emissivity
$I_{ss,peak}$	input current corresponding to peak incident flux, A	η_0	optical efficiency
$I_{ss,t}$	input current, A	θ	polar angle radians
k	conductivity, $\text{W m}^{-1} \text{K}^{-1}$	ρ_c	reflectivity
L	length of the receiver, m	σ	Stefan Boltzmann constant (5.67×10^{-8}), $\text{W m}^{-2} \text{K}^{-4}$
$\dot{m}$	mass flow rate, kg s^{-1}	<i>Subscripts</i>	
q	energy, W	ab	absorbed
$q_{ab,c}$	energy absorbed by the cavity, W	amb	ambient
$q_{ab,o}$	energy absorbed by outer side of the front plate, W	ap	aperture
$q_{ab,w}$	energy absorbed by the inner walls, W	bp	back plate
$q_{bm,t}$	incident irradiance using Bird's model, W m^{-2}	c	cavity
q_{cond}	conduction, W	$c,1$	cavity, inner
q_{conv}	convection, W	$c,2$	cavity, outer
q_{in}	incident energy, W	em	emitted
q^p	energy of each primary ray, W	es	escaped
q_{rerad}	energy emitted from the receiver walls, W	fp	front plate
q^s	energy of each secondary ray	g	gas
$q_{ss,peak}$	peak incident flux of the solar simulator, kW m^{-2}	in	inlet
$q_{ss,t}$	incident irradiance from solar simulator, kW m^{-2}	ins	insulation
R	resistance KW^{-1}	r	receiver
R_r	random number	ref	reflected
r	radius, m	x	cavity, front or back plate
T	temperature, K		

MCRT coupled with a heat transfer model is a method widely used to investigate the heat flux effect and the temperature distribution in solar applications. For example, MCRT and the Finite Volume Method (FVM) based software FLUENT is coupled to investigate the temperature distribution in a porous media receiver. User defined functions are introduced to solve solid–liquid phase change heat transfer problems [16–18] and chemical reaction kinetics [19]. MCRT and a three dimensional FVM have also been used to design and optimize parabolic trough based systems with receivers that have a heat transfer fluid [20] and a V-cavity absorber with rectangular fins [21]. Furthermore, the effect of having a glass cover over the reactor aperture on the heat flux distribution has also been investigated [22]. It is reported that an elliptic–circular cross-section of the glass cover can reduce thermal stress in a tube receiver, but the optimum aperture size is not obtained. An optimum aperture size with reference to an optimum operating temperature for any reactor geometry is reported [6]. However, the optimum aperture changes with the change in incident irradiation intensity [23] and with required operating temperature [24]. Therefore, a fixed optimum aperture with variable incident irradiation intensity over the course of a day cannot maintain a constant operating temperature due to its inability to adapt to fluctuating incident flux density.

To maintain quasi-constant internal cavity temperature in solar thermochemical applications, variable aperture mechanisms have been proposed and an aperture control device was patented in 1980 [25]. Several numerical investigations into the effect of aperture size on the temperature in a solar reactor cavity for thermochemical applications have been performed using Kirchhoff's law and assuming equal absorptivity and emissivity, [1,23,26–29]. No experimental work has been undertaken to date to validate the numerical model considering both heat transfer and optics and

employing different absorptivity and emissivity values. Also, no comparative analysis of fixed and variable apertures has been performed to quantify the increase in solar energy harvested.

Indoor experimental and numerical investigation of a thermochemical system requires a solar simulator. Different types of solar simulators (pulsed or continuous, single or multiple sun concentrations) with different light source types (e.g. metal halide lamp, xenon arc lamp etc.) have been used for indoor thermochemical investigations [30–34]. The basic configuration of a high flux solar simulator consists of an ellipsoidal reflector and a light source. The light source is often a xenon arc lamp [30,31] or metal halide lamp [32]. The center of the light source is placed on one foci of the ellipsoidal reflector while emitted light is concentrated on the second foci. A xenon arc lamp emits radiation from a small area between the cathode and anode. The radiation emission intensity is highest at the generation point, i.e. from the plasma ball at the tip of the cathode. Emission intensity decreases as it moves toward the anode [35]. Different arc designs for light source simulations have been employed, modeling the arc as a sphere or as a cylinder, [30,31,36,37]. This study proposes and validates the combination of a sphere and cylinder to represent the arc of the xenon arc lamp. A test that probably has not been performed yet.

In this research, extensive experimental work has been undertaken using a cylindrical solar receiver with different aperture diameters and varying incident irradiance intensity. The experimental results are used to validate a developed and modified in-house finite volume based coupled Monte Carlo/Heat Transfer (cMCHT) numerical model. The validated numerical model is used to investigate the optimum aperture size along with aperture variations required to maintain a constant temperature at higher input power.

2. Methodology

2.1. Experimental setup

The experimental setup consists of a high flux solar simulator, an insulated solar receiver, gas cylinders, thermocouples, mass flow controllers and a data acquisition system. The schematic illustration and snap shot of the experimental setup is shown in Fig. 1.

The solar receiver is made from Inconel Grade 601 and consists of a cylindrical cavity of 185.5 mm length, inner diameter of 140 mm, a wall thickness of 15 mm, a front plate and a back plate as shown in Fig. 1. The incident irradiance enters in the receiver through an aperture on the front plate. The aperture diameter can be changed by using front flanges of different inner diameters of 4 cm, 6 cm, 8 cm, 10 cm, 12 cm, and 14 cm. A transparent quartz window, 1/8 inches thick and 6 inches in diameter [38], is mounted between the front plate and the cavity. The cavity and the back plate of the receiver are insulated, as shown in Fig. 1(a), with flexible ceramic insulation [39] of ~8 cm thickness. Two stainless steel manifolds are used to feed the receiver cavity. The window manifold feeds the receiver through 4 'window ports' with an inclination of 30° toward the quartz window as shown in Fig. 1(a). This enables the cooling of the window. The cavity manifold feeds the receiver with 4 tangential inlets, 'cavity ports' in Fig. 1(a), to achieve vortex flow. All current experiments were conducted using inert gas i.e. Nitrogen gas (purity 99.999% – Grade 5.0) with a flow of 5 L min^{-1} ($1.67 \text{ m}^3 \text{ s}^{-1}$) from each manifold. An Omega FMA-A2411 [40] mass flow controller was used to control the flow in the window ports while a MKS M100B [41] mass flow controller was used to control the flow in the cavity ports. A single port at the back plate serves as the receiver exhaust. The cavity temperatures were measured using K type thermocouples [40] spaced axially and radially along the receiver body to obtain the temperature profile in the cavity. In order to ensure no direct exposure of the thermocouples to the incident irradiance, they were placed at an offset distance of 5 mm from the inner walls of the cavity. One K-type thermocouple was placed at the exhaust to record outlet gas temperature. The voltage output of the thermocouples was measured using thermocouple input module NI 9213 [42]. All data were recorded at a sample rate of 1 Hz and collected through LabVIEW software [43] until steady state temperatures were achieved (steady state criteria: less than 0.6 K change per hour). The steady state temperatures were compared with numerical model results and are reported in Section 3.1.

A commercially available Xenoluxe Lamphouse, [44] is used as a high flux solar simulator. The solar simulator employs a 7 kW xenon short arc lamp [45] and an ellipsoidal reflector. A detailed description and characterization of the high flux solar simulator has been undertaken and reported by authors' elsewhere [46]. The simulator's ellipsoidal reflector has a focal point 55 mm from the vertex on the optical axis. The second focal point is 825 mm from the first focal point on the optical axis. The outer truncated diameter is 340 mm. The light source is placed at the first focal point and the receiver aperture is located at the second focal point. The flux distribution and cumulative power of the solar simulator is adjustable by changing the input current. Input current can be adjusted from 110 A to 153 A, resulting in a peak flux variation in the range of $2074\text{--}3811 \text{ kW m}^{-2}$ [46]. The cumulative power incident at an aperture diameter of 70 mm varies in the range of $709\text{--}960 \text{ W}$. This is correlated with an actual incident irradiance via the Bird and Hulstrom (1981) model [47]. The reference day for this study is July 1, 2011 at 39.74N , 105.18W and at an elevation of 1829 m [48]. The irradiance is correlated with the input current of the solar simulator through fitting the data of the selected reference day. The correlation between current of the solar simulator and incident irradiance is shown in Eq. (1).

$$I_{ss,t} = \frac{q_{bm,t}}{6.695454546} \quad (1)$$

where $I_{ss,t}$ represents input current of the solar simulator and $q_{bm,t}$ represents incident irradiance found using Bird's model, [47] at a given time of the day.

The input current of the solar simulator is related to the incident irradiance available at the second focal point of the solar simulator using the following equation:

$$q_{ss,t} = \frac{q_{ss,peak}}{(I_{ss,peak}/I_{ss,t})} \quad (2)$$

where $q_{ss,t}$ is the incident irradiance from the solar simulator at a given time of the day, $q_{ss,peak}$ is the peak incident flux of the solar simulator, and $I_{ss,peak}$ is the input current corresponding to peak incident flux.

The comparison between the reference incident solar irradiance using Bird's model to the irradiance available at the front plane is shown in Fig. 6. Three specific power levels, 110 A, 125 A and 153 A, have been selected to represent early morning (06:30), morning (07:15) and noon (12:00) time irradiance for the reference day respectively. Finally, an experiment has been performed

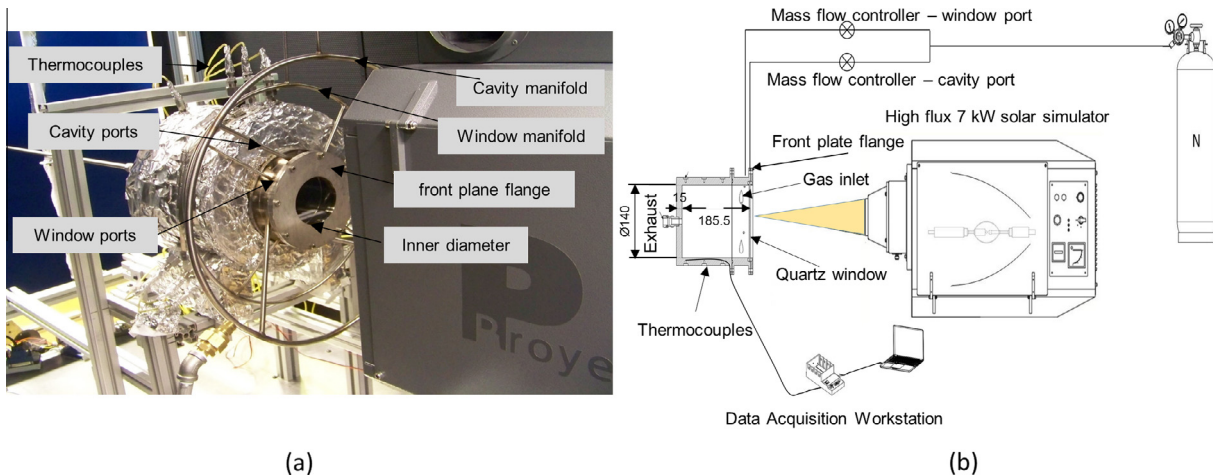


Fig. 1. Snapshot and schematic illustration of the experimental setup (a) the solar reactor setup and (b) the overall setup.

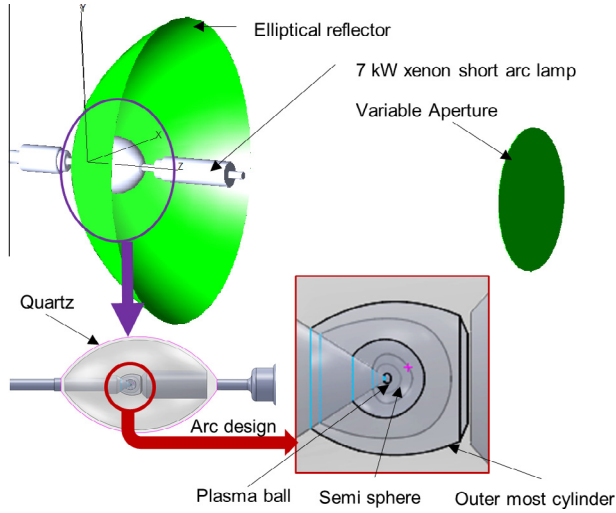


Fig. 3. Schematic illustration of concentrating system created in TracePro®.

where q^p is the energy of each primary ray, q_s is the total incident energy and N_{rays} is the total number of rays. The probability of being reflected or absorbed is calculated using a uniformly distributed random number R_r between zero and one. For a surface with reflectivity of ρ_c , the probability is calculated as, [19]:

$$\begin{aligned} R_r < \rho_c, & \text{ reflection} \\ R_r > \rho_c, & \text{ absorption} \end{aligned} \quad (4)$$

The reflection direction of the incident primary rays follows the Fresnel law of optics. The energy of secondary rays, q^s is calculated as:

$$q^s = \frac{q_{rerad}}{N_{rays}} \quad (5)$$

Energy emitted, q_{rerad} from the receiver is obtained using Eq. (8). To obtain the direction of secondary ray, the polar and cone angle, θ and φ respectively, are calculated via the following equation:

$$\begin{aligned} \theta &= \sin^{-1} \sqrt{R_r} \\ \varphi &= 2\pi R_r \end{aligned} \quad (6)$$

The fate of each ray is tracked until it is absorbed by the walls or escapes through the aperture or the exhaust. The MC Ray tracing method is coupled with the finite volume heat transfer algorithm to model steady state conservation of energy in the receiver:

$$\begin{aligned} q_{ab,c} + q_{ab,w} + q_{es} + q_{ab,o} - q_{rerad} - q_{conv} - q_{cond} \\ - \dot{m}_g c_p (T_g - T_{in}) \\ = 0 \end{aligned} \quad (7)$$

where $q_{ab,c}$ is the energy absorbed by the cavity, $q_{ab,w}$ is the energy absorbed by the inner walls (front plate and back plate), q_{es} is the energy escaped from the receiver, $q_{ab,o}$ is the energy absorbed by the outside of the front plate, and q_{rerad} is the energy emitted from the receiver walls. q_{conv} is the energy transferred due to convection to the adjacent gas phase, q_{cond} is the energy transferred to the receiver by conduction, $\dot{m}_g$ is the mass flow rate of the gas, c_p is the heat capacity of the gas, T_g is the temperature of the gas and T_{in} is the inlet temperature of the gas.

For the receiver's cavity and walls, the emitted energy q_{rerad} is calculated as, [4]:

$$q_{rerad} = A_x \epsilon \sigma (T_x^4 - T_{amb}^4) \quad (8)$$

where A is the area and the subscript x represents either the cavity or wall. The emissivity is denoted by ϵ , σ is the Stefan Boltzmann constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$) and T represents the temperature with the subscript x denoting either the wall or the cavity. T_{amb} is the ambient temperature.

The energy transferred through convection is calculated as:

$$q_{conv} = h_c A_{c,1} (T_c - T_g) \text{ with } A_{c,1} = 2\pi r_{c,1} w_c \quad (9)$$

where h_c denotes the convection heat transfer coefficient. $A_{c,1}$ is the inner surface area of the cavity, T_c is the temperature of the cavity, T_g is the temperature of the gas, $r_{c,1}$ is the inner radius of the cavity, and w_c is the width of the cavity.

The energy transferred due to conduction q_{cond} is calculated as:

$$q_{cond} = \frac{T_x - T_{amb}}{R_x} \quad (10)$$

where T_x represents the temperature of the walls and the cavity, T_{amb} is the ambient temperature and R_x denotes the thermal resistance of the walls and the cavity.

The thermal resistance does not depend on the conductivity of the materials alone, but is also a function of the geometry. The thermal resistance for different segments of the receiver, referred to as cavity, front plate and back plate, is calculated as:

Cavity:

$$R_c = \frac{1}{h_{out} A_{c,2}} + \frac{\ln\left(\frac{r_{c,2}}{r_{c,1}}\right)}{2\pi k_r L} + \frac{\ln\left(\frac{r_{c,2} + t_{ins}}{r_{c,2}}\right)}{2\pi k_{ins} L} \quad (11)$$

$$\text{where } A_{c,2} = 2\pi(r_{c,2} + t_{ins})w_c$$

Wall, front plate:

$$R_{fp} = \frac{1}{h_{out} A_{fp}} \quad (12)$$

$$\text{where } A_{fp} = 2\pi(r_{c,1} - r_{ap})w_{fp}$$

Wall, back plate:

$$R_{bp} = \frac{1}{h_{out} A_{bp}} + \frac{t_{bp}}{k_r A_{bp}} + \frac{t_{ins}}{k_{ins} A_{bp}} \quad (13)$$

$$\text{where } A_{bp} = 2\pi(r_{c,1} - r_{bp})w_{bp}$$

In Eqs. (11)–(13), $A_{c,2}$, A_{fp} and A_{bp} are the area of the outer surface of the cavity, the front plate area and the back plate area respectively. w_c , w_{fp} and w_{bp} are the corresponding widths. $r_{c,2}$ is the outer surface radius of the cavity. The thermal conductivity of the receiver and insulation material are denoted as k_r and k_{ins} respectively. L is the length of the receiver. The thickness of the back plate and insulation is represented as t_{bp} and t_{ins} respectively while h_{out} is the heat transfer coefficient.

The temperatures of the gas and different segments of the receiver in Eqs. (7)–(10) are the unknowns and coupling is required to satisfy the energy balance equation. The absorption and escape terms are calculated using the MC Ray tracing method. The temperature of the gas is calculated using a steady state energy balance for the gas only, and given as:

$$q_{conv} - \dot{m}_g c_p (T_g - T_{in}) = 0 \quad (14)$$

Eq. (7) can be re-written for different segments of the receiver after incorporating Eqs. (8)–(10) using the appropriate thermal resistances from Eqs. (11)–(13). By separating equations for different segments of the receiver, fourth-order equations are obtained which are solved simultaneously with equation 14 to obtain the unknown temperatures. Then, the calculated temperatures are used to check the energy balance satisfaction.

The optical efficiency, defined as the ratio of radiation absorbed by the receiver to the incident radiation at the aperture [52], is calculated using Eq. (15).

$$\eta_o = \left(\frac{q_{ab,c} + q_{ab,w}}{q_{in}} \right) \quad (15)$$

The power ratio is calculated to quantify the relative effect of the incident power, q_{in} , changes during the day and the change in aperture size (Fig. 6). The power ratio is defined as the power incident at the selected aperture size to the total power incident at the receiver. The comparison of variation in optical efficiency and power ratio with the change in aperture size and the time of the day is presented in Section 3.2.

2.3. Numerical model input properties

The schematic illustration of inputs, outputs and variable parameters of the cMCHT model is shown in Fig. 4. The required input of this model are incident solar radiation, geometry and gas mass flow rate in addition to the material/surface dependent properties such as heat capacity, conductivity, absorptivity and emissivity.

The geometry is discretized using the finite volume method with the following assumptions:

- Interior surfaces of the receiver are considered diffuse-gray.
- The aperture opening is considered to be a perfectly transmitting medium; i.e. no reflectance or absorptance takes place inside the medium.
- The gas phase is considered to be a non-participating medium.
- Conduction is assumed only in the radial direction i.e. one dimensional heat transfer.

The schematic illustration of the control volumes of the receiver and the gas phase is shown in Fig. 5.

To determine the control volume's size, a grid independence study was carried out and results are shown in Section 2.4. Finally, the cavity of the solar receiver (the gas) is discretized into 235 equal width control volumes while the back plate and front plate of the receiver are discretized using 100 and 50 control volumes respectively. The other input properties for performing thermal analysis are tabulated in Table 1.

The model calculates the temperature of the different segments of the receiver; i.e. the front plate, the back plate and the cavity. These results are used to validate the numerical model in comparison to the experimental measurements. Then, the validated model is used to simulate a new scenario with a bigger light source (30 kW) in order to investigate the optimum aperture size and

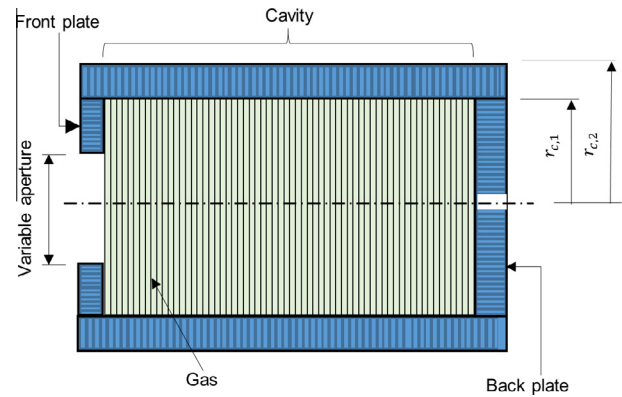


Fig. 5. Schematic illustration of control volumes for the reactor and the gas phase with notational representation of radius.

Table 1

Input properties for the thermal analysis using the coupled Monte Carlo/heat transfer model.

Properties	Units	Input values
Volume flow rate, $\dot{m}_g$	(L min ⁻¹)	10
Heat capacity of nitrogen, c_p	(J kg ⁻¹ K ⁻¹)	1056
Stefan Boltzmann constant, σ	(W m ⁻² K ⁻⁴)	5.67×10^{-8}
Inlet gas temperature/ambient temperature, T_{in}	(K)	295
Heat transfer coefficient of air, h_{out}	(W m ⁻² K ⁻¹)	5
Thermal conductivity of Inconel, k_{dl}	(W m ⁻¹ K ⁻¹)	20.9
Thermal conductivity of insulation, k_{ins}	(W m ⁻¹ K ⁻¹)	1
Thermal conductivity of air, k_g	(W m ⁻¹ K ⁻¹)	0.05
Emissivity, ε		0.85

the possibility of maintaining a constant temperature in the range of 1000–1200 K via a variable aperture.

2.4. Grid independence studies

For the grid independency study, the cavity of the receiver was sub-divided into 200–600 different equal sized control volumes (2 and 3 times the final selected size). Simulations were performed and the average steady state temperatures of the cavity were 700.60 K, 700.59 K, 700.63 K, 700.59 K, 700.54 K and 700.58 for 200, 235, 300, 400, 500 and 600 total volumes respectively. The maximum temperature difference of 0.09 K shows that the selected size and thus the solution satisfies the grid independence condition. Obviously the number of volumes could be further reduced, but there was no significant improvement to the calculation time.

3. Results and discussions

3.1. Validation

This section describes the validation of both the optical and heat transfer (cMCHT) models. For the optical model, the measured incident heat flux using the solar simulator is compared with the numerical values in Fig. 6(a). Simulated and experimental power variation over a day in addition to Bird's model is shown in Fig. 6(b). An excellent match has been achieved with a maximum variation of ~2.2%.

Table 2 presents the total number of rays entering through the aperture and the corresponding power for each selected time of the day and selected aperture diameter.

To validate the cMCHT model, the calculated average steady state temperatures of the cavity at selected apertures and

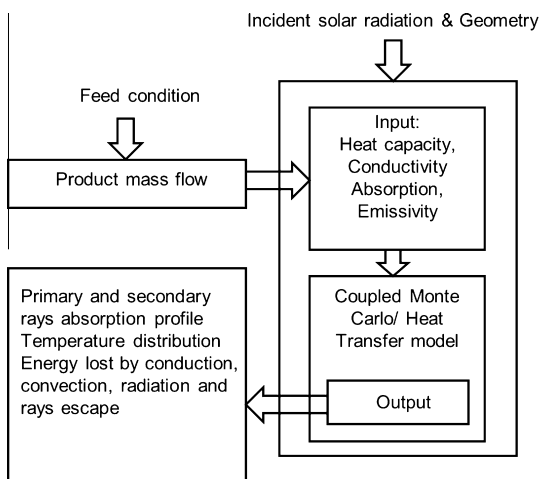


Fig. 4. Schematic illustration of inputs, output and variable parameters of coupled Monte Carlo/heat transfer model.

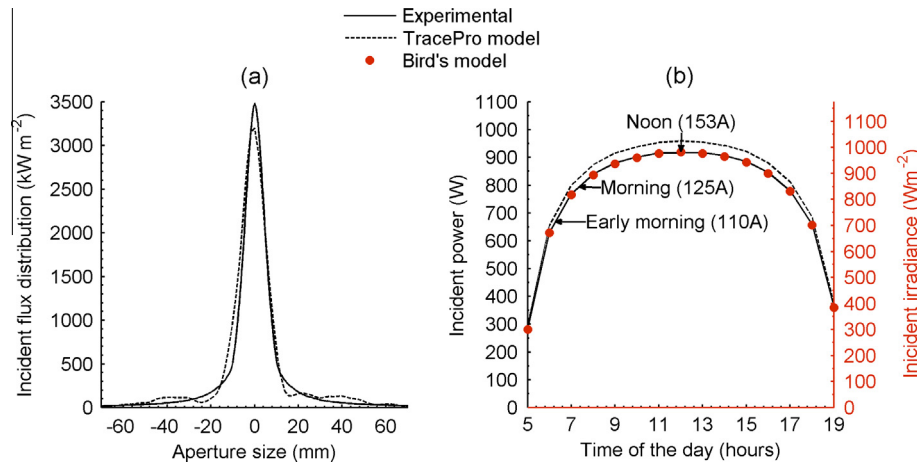


Fig. 6. Comparison of experimental and numerical (a) flux distribution over an aperture diameter of 14 cm and (b) hourly incident power/irradiance at a target surface of 7 cm diameter.

Table 2

Number of rays entering through the aperture diameter and corresponding power at selected time of the day i.e. 06:30, 07:15 and 12:00.

Aperture diameter (cm)	4	6	8	10	12	14
Number of rays	256,530	320,351	401,278	479,023	531,462	566,977
Power at 0600 (W)	481	601	753	898	997	1064
Power at 0715 (W)	581	726	909	1085	1204	1284
Power at 1200 (W)	755	943	1181	1409	1564	1668

irradiance levels were compared with the corresponding experimental values.

The comparison is shown in Fig. 7 and an excellent agreement is observed with an average difference of $\pm 0.2\%$. Consequently, the developed numerical model is considered valid and able to predict cavity temperatures at different configurations of apertures and input power.

3.2. Prediction of thermal behavior at higher input power

The predicted and measured cavity temperatures in Fig. 7 show the expected significant variation throughout the day. This also shows that there is a potential to control the temperature by varying the aperture size. Therefore, a new simulation scenario is selected to explore the design aspects of achieving higher and constant cavity temperatures in the range of 1000–1200 K. Preliminary

calculations estimated the need for a xenon short arc lamp light source of 30 kW with an electrical to radiation conversion efficiency of 50%, [53]. Using the developed model, a new light source is modeled again. Fig. 8 presents the good agreement between the modeled incident power with the irradiance obtained from Bird's model using the same reference date and location – July 1, 2011 at NREL).

Fig. 8 shows that the incident power at 05:00 is only 2000 W, but at 12:00 it reaches a peak of 6027 W, before it is reduced down to 2500 W at 19:00. The variation from 05:00 to 12:00 is $\sim 66\%$ and $\sim 58\%$ from 12:00 to 19:00. This variability poses a significant challenge of maintaining constant temperature in the solar receiver. The power and flux distribution at aperture diameters of 4 cm, 6 cm, 8 cm, 10 cm, 12 cm and 14 cm at 06:30 (early morning), 07:15 (morning) and 12:00 (noon) are shown in Fig. 9.

In conjunction with the thermal analysis described earlier, Fig. 9 also shows that knowing the incident power changes during the

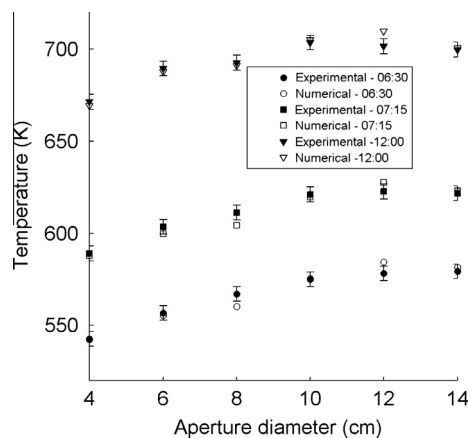


Fig. 7. Comparison of experimental and numerical steady state temperatures at different aperture diameters and irradiance levels at early morning, morning and noon time.

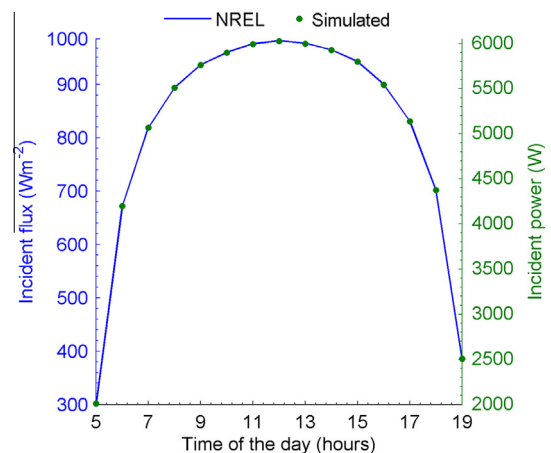


Fig. 8. Comparison of the modeled incident solar irradiance between the developed optical model (30 kW xenon short arc lamp) and Bird's model (July 1, 2011 at NREL).

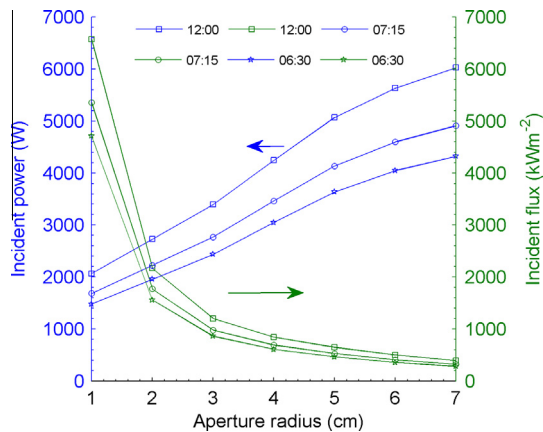


Fig. 9. Incident power and flux at different aperture diameters at selected time of the day.

day for each aperture size is not enough to maintain a constant temperature inside the cavity. This is expected because of the multiple and “competitive” phenomena. Therefore, a full thermal analysis has to be performed for each case varying the time of the day and aperture size in order to estimate cavity temperatures. These calculations are summarized in Fig. 10(a).

The same figure also reveals that there is no single optimum aperture for the whole day; or for any irradiance level. For example, an aperture diameter of 10 cm results in maximum temperature during the early and late hours of the day (06:00 to 09:00 and from 16:00 to 18:00). For the remainder of the day maximum temperature is achieved with an aperture diameter of 8 cm. The reasons for this temperature variation are investigated by calculating optical efficiency using Eq. (15) and the re-radiation losses for selected aperture sizes between 06:00 to 12:00. These results are compared with the power ratio and shown in Fig. 11.

It is found that the power ratio is independent of the time of the day and increases with the increase in aperture diameter. This analysis also shows that the optical efficiency decreases with the increase in the aperture diameter while power ratio and radiation losses show opposite behavior. It is evident from Fig. 11 that the re-radiation losses start to dominate beyond an aperture diameter of 8 cm. In this case, above 8 cm aperture diameter, increases in power ratio do not yield higher temperatures as shown previously in Fig. 10.

Although, optical efficiency is higher for aperture diameters from 2 cm to 6 cm (>64%), the significant decrement in power ratio (<56%) results in lower receiver temperature. The difference between the maximum and the minimum cavity temperature

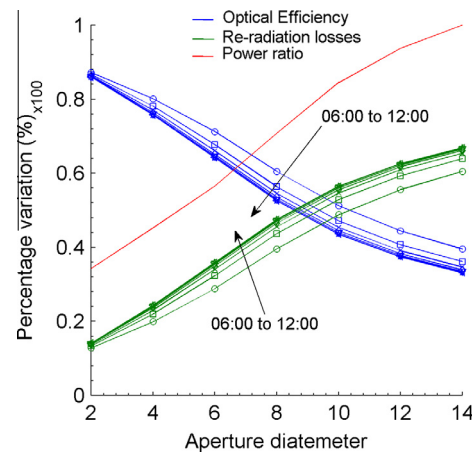


Fig. 11. Optical efficiency, re-radiation losses and power ratio with the change in the aperture diameter and the time of the day.

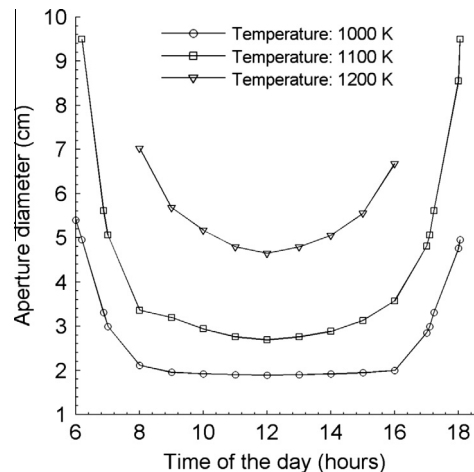


Fig. 12. Aperture diameters to maintain constant temperature of 1000 K, 1100 K and 1200 K over the day.

(see Fig. 10(b)) is estimated to be between 190 K and 170 K for the aperture diameters of 8 cm and 10 cm respectively. This variability is significant when it comes to receiver performance. In the case of direct methane cracking, a 190 K change in temperature could result in ~5–55% variation in hydrogen production [2].

On the other hand, varying the aperture diameter could be used to maintain a constant temperature in the receiver cavity. This was

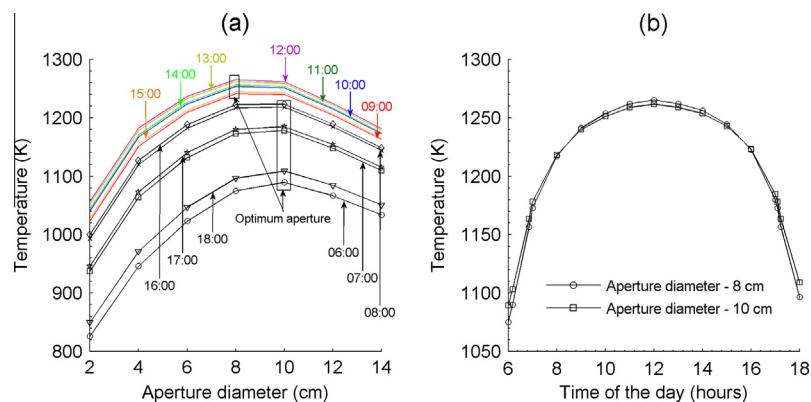


Fig. 10. Steady state cavity temperatures vs (a) the different aperture diameters for each time of the day and (b) vs time of the day for aperture diameter of 8 cm and 10 cm.

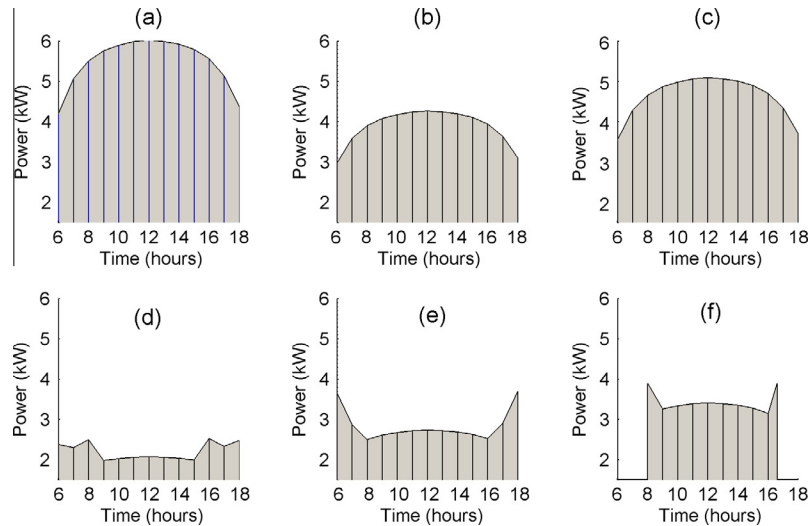


Fig. 13. Solar reactor power plots vs time of the day for (a) incident, (b) intercepted for 8 cm aperture, (c) intercepted for 10 cm aperture, (d) intercepted while maintaining constant temperature of 1000 K, (e) intercepted while maintaining constant temperature of 1100 K and (f) intercepted while maintaining constant temperature of 1200 K.

Table 3
Power consumption and power savings for fixed and variable aperture.

		Fixed aperture			Variable aperture		
		14 cm	10 cm	8 cm			
		<i>Temperature variations with peak</i>			<i>Maintain temperature at</i>		
		(1182 K)	(1262 K)	(1265 K)	1000 K	1100 K	1200 K ^a
Power consumption	(Peak) kW day	148 K	172 K	190 K	26.8	33.2	26.9
		66.9	56.6	47.4			
<i>Power savings relative to aperture of</i>							
14 cm	kW day	0	10.4	19.6	40.2	33.8	40.1
10 cm	kW day		0	9.2	29.8	23.4	29.7
8 cm	kW day			0	20.6	14.2	20.5

^a For 8.75 h and not 12 h like the rest of the cases.

further investigated for this simulation scenario across three different temperatures: 1000 K, 1100 K and 1200 K.

The results of this investigation are presented in Fig. 12 and show that a constant temperature of

- 1000 K can be maintained for 12 h for the selected day by varying the aperture diameter in the range of 5.4 cm to 1.9 cm.
- 1100 K can be maintained for 11 h 52 min by varying aperture diameter in the range of 2.7–9.5 cm.
- 1200 K can only be maintained for 8 h by varying aperture diameter in the range of 4.6–7 cm.

This analysis shows that a variable aperture is capable of maintaining constant temperature in a solar thermal receiver over the day and thereby improving the efficiency of the system. An efficiency which otherwise is significantly affected by the varying receiver temperature.

Another critical aspect for the comparison of variable or fixed apertures, is the overall solar power utilization and savings. Figs. 10–12 provide information about the steady-state temperatures of the cavity over the day, as well as optical efficiency and re-radiation losses, but do not provide any information about overall power utilizations/savings. Therefore, the incident and intercepted power at the aperture is calculated over a day for the current simulation scenario (Fig. 13).

The total daily consumed power is calculated by integrating the power plots in Fig. 13, using standard numerical tools [54].

Table 3 summarizes the power consumptions for the fixed and variable aperture combinations. For example, a fixed aperture of 8 cm will “consume” 47.4 kW day and the temperature will vary from ~1050 K to 1250 K. However, in order to maintain a temperature of 1100 K the receiver needs only 33.2 kW day while the aperture size will vary from ~10 cm to 3 cm. Similarly, the fixed aperture diameter of 14 cm consumes 60%, 50% and 60% more power compared to variable apertures, to maintain constant temperature of 1000 K, 1100 K and 1200 K, respectively. Note, that in the current scenario a temperature of 1200 K can only be maintained for 8.75 h. This analysis shows that a variable aperture utilizes less than half of the total incident power. Furthermore, the variable aperture operates below an aperture diameter of 6 cm for 12 h, 11 h and 6.6 h to maintain a constant temperature of 1000 K, 1100 K and 1200 K. As a result, re-radiation losses stay below 36% as shown in Fig. 10. Hence, it can be concluded that the variable aperture harvests solar energy more efficiently than a fixed aperture. This also means that more than half of the incident energy on the solar simulator is lost as it is not intercepted by the variable aperture. Therefore, further research and modifications in the optics are required to increase the energy interception by the variable aperture. Optical losses and the effect of the quartz window's shape and thickness on thermal performance of the

receiver can be improved. Some characteristic examples of such modification have already been reported in the introduction, [8,11–13,22].

4. Conclusions

The present work, experimentally and numerically studied the effects of the aperture size on receiver cavity temperatures. Extensive experimental work was undertaken using a cylindrical solar receiver with aperture diameters of 4 cm, 6 cm, 8 cm, 10 cm, 12 cm and 14 cm at varying incident irradiance intensity in order to mimic the sunlight intensity at early morning (06:30), morning (07:15) and noon (12:00) time on the selected reference day. A high flux solar simulator, consisting of a 7 kW xenon short arc lamp, was employed as a light source while variation in irradiance was simulated by changing the input current. Based on the experimental setup, an optical and a thermal analysis model was developed and validated. The optical model was simulated using the commercial software TracePro®, while the thermal analysis model was based on a Monte Carlo/finite volume heat transfer model. Upon the successful validation of the models, a new scenario was simulated with a more brilliant light source (30 kW) in order to investigate the optimum aperture size and the possibility to maintain constant temperatures in the range of 1000–1200 K. The following conclusions can be made:

- (1) The optical model is a useful tool to simulate a high flux solar simulator. Very good agreement between experimental and numerical results is found with a maximum variation of 2.2%.
- (2) The thermal analysis model achieved excellent agreement with the experimental results with an average temperature difference of $\pm 0.2\%$.
- (3) The optimum aperture size changes with the change in irradiance intensity and therefore time of the day.
- (4) The difference between maximum and minimum temperature was found to be in the range of 170–190 K for fixed aperture diameters between 8 cm and 10 cm. This result is rather significant from a reaction control point of view.
- (5) Variable apertures can maintain constant temperature in a cylindrical solar receiver. In doing so the variable aperture consumes almost half the power of a fixed aperture.

Overall, this study concludes that the developed model has been successfully validated and can further be used to assess and design more efficient systems. As an example, a thorough analysis showed that the application of a variable aperture can maintain constant temperature in a cylindrical solar cavity receiver over a day. This also results in a decrease in power consumption compared to a fixed aperture. However, better energy interception is necessary to exploit the loss of energy when using a variable aperture.

Acknowledgements

This publication was made possible by a NPRP award [NPRP 6 – 116 – 2 – 044] from the Qatar National Research Fund (a member of The Qatar Foundation). The statements made herein are solely the responsibility of the authors. The HPC resources and services used in this work were provided by the IT Research Computing group in Texas A&M University at Qatar. IT Research Computing is funded by the Qatar Foundation for Education, Science and Community Development (<http://www.qf.org.qa>).

The authors would like to express their gratitude to Dr. Nesrin Ozalp, who was the founder and until recently director of the

Sustainable Energy Research Lab, where this work has been conducted, and one of the initial investigators of the aforementioned award.

References

- [1] Ozalp N et al. A smart solar reactor for environmentally clean chemical processing. *Chem Eng Trans* 2011;25:989–94.
- [2] Abánades A et al. Experimental analysis of direct thermal methane cracking. *Int J Hydrogen Energy* 2011;36(20):12877–86.
- [3] Dahl JK, Weimer AW, Krantz WB. Sensitivity analysis of the rapid decomposition of methane in an aerosol flow reactor. *Int J Hydrogen Energy* 2004;29(1):57–65.
- [4] Charvin P et al. Analysis of solar chemical processes for hydrogen production from water splitting thermochemical cycles. *Energy Convers Manage* 2008;49(6):1547–56.
- [5] Palumbo R et al. Reflections on the design of solar thermal chemical reactors: thoughts in transformation. *Energy* 2004;29(5–6):727–44.
- [6] Steinfeld A, Schubnell M. Optimum aperture size and operating temperature of a solar cavity-receiver. *Sol Energy* 1993;50(1):19–25.
- [7] Haueter P et al. The production of zinc by thermal dissociation of zinc oxide—solar chemical reactor design. *Sol Energy* 1999;67(1–3):161–7.
- [8] Kogan A, Kogan M. The tornado flow configuration—an effective method for screening of a solar reactor window. *J SolEnergy Eng* 2002;124(3):206–14.
- [9] Hirsch D, Steinfeld A. Solar hydrogen production by thermal decomposition of natural gas using a vortex-flow reactor. *Int J Hydrogen Energy* 2004;29(1):47–55.
- [10] Rodat S, Abanades S, Flamant G. Co-production of hydrogen and carbon black from solar thermal methane splitting in a tubular reactor prototype. *Sol Energy* 2011;85(4):645–52.
- [11] Yeheskel J, Epstein M. Thermolysis of methane in a solar reactor for mass-production of hydrogen and carbon nano-materials. *Carbon* 2011;49(14):4695–703.
- [12] Steinfeld A, Fletcher EA. A solar receiver–reactor with specularly reflecting walls for high-temperature thermoelectrochemical and thermochemical processes. *Energy* 1988;13(4):301–11.
- [13] Steinfeld A, Fletcher EA. Solar energy absorption efficiency of an ellipsoidal receiver–reactor with specularly reflecting walls. *Energy* 1988;13(8):609–14.
- [14] Shuai Y, Xia X-L, Tan H-P. Radiation performance of dish solar concentrator/cavity receiver systems. *Sol Energy* 2008;82(1):13–21.
- [15] Mao Q, Shuai Y, Yuan Y. Study on radiation flux of the receiver with a parabolic solar concentrator system. *Energy Convers Manage* 2014;84:1–6.
- [16] Wang F et al. Heat transfer analyses of porous media receiver with multi-dish collector by coupling MCRT and FVM method. *Sol Energy* 2013;93:158–68.
- [17] Wang F et al. Thermal performance analysis of porous media receiver with concentrated solar irradiation. *Int J Heat Mass Transf* 2013;62:247–54.
- [18] Fuqiang W et al. Thermal performance analysis of porous medium solar receiver with quartz window to minimize heat flux gradient. *Sol Energy* 2014;108:348–59.
- [19] Wang F et al. Numerical analysis of hydrogen production via methane steam reforming in porous media solar thermochemical reactor using concentrated solar irradiation as heat source. *Energy Convers Manage* 2014;87:956–64.
- [20] Wu Z et al. Three-dimensional numerical study of heat transfer characteristics of parabolic trough receiver. *Appl Energy* 2014;113:902–11.
- [21] Xiao X et al. Experimental and numerical heat transfer analysis of a V-cavity absorber for linear parabolic trough solar collector. *Energy Convers Manage* 2014;86:49–59.
- [22] Fuqiang W et al. Effects of glass cover on heat flux distribution for tube receiver with parabolic trough collector system. *Energy Convers Manage* 2015;90:47–52.
- [23] Abdulla S, et al. Design, manufacturing and testing of a camera-like aperture mechanism for a solar reactor. In: ASME 2011 international mechanical engineering congress and exposition. Denver, Colorado, USA: ASME; 2011. p. 305–19.
- [24] Tescari S et al. Thermal model for the optimization of a solar rotary kiln to be used as high temperature thermochemical reactor. *Sol Energy* 2013;95:279–89.
- [25] Jubb A. Solar heat aperture control apparatus. In: Patent No. US4222367 A1980. Unites States: Rolls-Royce Limited. p. 5.
- [26] Al Hamidi Y, et al. Design, manufacturing and testing of an aperture mechanism for a solar reactor. In: 5th International conference on energy sustainability. Washington, DC: ASME; 2011.
- [27] Menon A, Ozalp N. Optical analysis of variable aperture mechanism for a solar reactor. *World Acad Sci Eng Technol* 2011;59:925–9.
- [28] Ozalp N et al. Effect of Cameralike aperture in quest for maintaining quasi-constant radiation inside a solar reactor. *J Mech Des* 2011;133(2):021002.
- [29] Usman S, Ozalp N. Numerical and optical analysis of weather adaptable solar reactor. In: 9th International conference on heat transfer, fluid mechanics and thermodynamics (HEFAT 2012), Malta; 2012.
- [30] Krueger KR. Design and characterization of a concentrating solar simulator. Graduate school, University of Minnesota; 2012.
- [31] Petrasch J et al. A novel 50 kW 11,000 suns high-flux solar simulator based on an array of xenon arc lamps. *J SolEnergy Eng* 2006;129(4):405–11.

- [32] Codd DS et al. A low cost high flux solar simulator. *Sol Energy* 2010;84(12): 2202–12.
- [33] Kuhn P, Hunt A. A new solar simulator to study high temperature solid-state reactions with highly concentrated radiation. *Sol Energy Mater* 1991; 24(1–4):742–50.
- [34] Dibowski H-G. High-flux solar furnace and xenon-high-flux solar simulator <http://www.dlr.de/sf/en/desktopdefault.aspx/tabid-8558/14717_read-28267/>; 2013 [13.06.13].
- [35] OSRAM. XBO® – theater lamps. Technology and application. Munich: OSRAM GmbH; 2014. p. 48.
- [36] Wang W et al. Design and validation of a low-cost high-flux solar simulator using Fresnel lens concentrators. *Energy Proc* 2014;49:2221–30.
- [37] Costandy J et al. Effect of reactor geometry on the temperature distribution of hydrogen producing solar reactors. *Int J Hydrogen Energy* 2012;37(21): 16581–90.
- [38] GM-Associates. Quartz window <<http://gmassoc.thomasnet.com/item/quartz-discs-2/quartz-polished-discs/7500-347/>>; 2014 [28.04.14].
- [39] Lydall. Lytherm® ceramic fiber papers. Lydall – Rochester, NH 03866: Rochester; 2001. p. 4.
- [40] Omega <http://www.omega.co.uk/pptst/FMAA2100_2200_2300_2400.html>; 2014 [28.04.14].
- [41] MKS-Instrument <<http://www.mksinst.com/product/product.aspx?ProductID=72>>; 2014 [28.04.14].
- [42] National-Instruments. NI 9213 <<http://sine.ni.com/nips/cds/view/p/lang/en/nid/208788>>; 2014.
- [43] National-Instruments <<http://www.ni.com/labview/>>; 2014 [April 2014].
- [44] Proyecson SA. Lamphouse Xenoluxe XL-7000 W <<http://en-proyecson.ecran.es/products/instalaciones-audiovisuales/productos/pantallas-hinchables/airscreen.html>>; 2013 [25.07.13].
- [45] OSRAM. XBO 7000 W/HS XLOFR <<http://www.osram.com/osram.com/products/lamps/specialty-lamps/xbo/xbo-xtreme-life/index.jsp>>; 2013 [12.06.13].
- [46] Sarwar J et al. Description and characterization of an adjustable flux solar simulator for solar thermal, thermochemical and photovoltaic applications. *Accepted Sol Energy* 2013.
- [47] Bird RE, Hulstrom RL. A simplified clear sky model for direct and diffuse insolation on horizontal surfaces. *Solar Energy Res Inst* 1981.
- [48] NRELdatabase. Solar radiation research laboratory, daily plots and raw data files <http://www.nrel.gov/midc/srnl_rsp2/>; 2014 [15.06.14].
- [49] Howell JR, Siegel R, Menguc MP. Thermal radiation heat transfer. 5th ed. New York: CRC Press, Taylor & Francis group; 2011.
- [50] Lambda Research Corporation. TracePro <<http://www.lambdaresearch.com/tracepro/>>; 2014 [05.01.14].
- [51] Xiao G et al. Optical and thermal performance of a high-temperature spiral solar particle receiver. *Sol Energy* 2014;109:200–13.
- [52] Balghouthi M et al. Optical and thermal evaluations of a medium temperature parabolic trough solar collector used in a cooling installation. *Energy Convers Manage* 2014;86:1134–46.
- [53] Krueger KR, Davidson JH, Lipiński W. Design of a new 45 kWe high-flux solar simulator for high-temperature solar thermal and thermochemical research. *J SolEnergy Eng* 2011;133(1):011013.
- [54] MathWorks. Trapezoidal numerical integration <<http://www.mathworks.com/help/matlab/ref/trapz.html>>; 2014 [20.08.14].